

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	S.Roshni	Reg. No.:	192525185
Section / Batch:	AI&ML	Date:	20/07/2026

Code of Conduct

I S.Roshni(192525185)_____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: S.Roshni_____

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.



Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:		100 Marks	



Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

31	192525185	SHAIK ROSHNI	1. Ten documents are requested for printing. Illustrate the mechanism of printer operations. Consider the document number as 1003, 1008, 1010, 1005, 1009, 1004. Identify the suitable data structure and draw the diagram to print the documents. 2. The university announced the selected candidates register number for placement training. The student XXX, Register Number 20142010 wishes to check whether his name is listed or not. The list is not sorted in any order. Identify the searching technique that can be applied and explain the searching steps with suitable procedure. List includes 20142015, 20142033, 20142011, 20142017, 20142010, 20142056, 20142003
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Course Name: Data Structure

CO2 AT1

S. Roshni

Reg No: 192525185
Course Code: CSA0305

1) Printer Operations

Problem

Documents waiting for printing:

1003, 1008, 1010, 1005, 1009, 1004

Suitable Data Structure

Queue (FIFO - First in First out)

- The first document entered is printed first
- New documents are inserted at Rear
- Printed documents are removed from the Front

Queue Diagram

Front

1003	1008	1010	1005	1009	1004	
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Printing process

Step 1:

Print 1003

Front

1008	1010	1005	1009	1004
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rear

Step 2:

Print 1008

Remaining Queue

1010 → 1005 → 1009 → 1004

Step 3:

Print 1010

Remainin Queue

1005 → 1009 → 1004

Step 4:

Print 1005

1009 → 1004

Step 5:

Print 1009

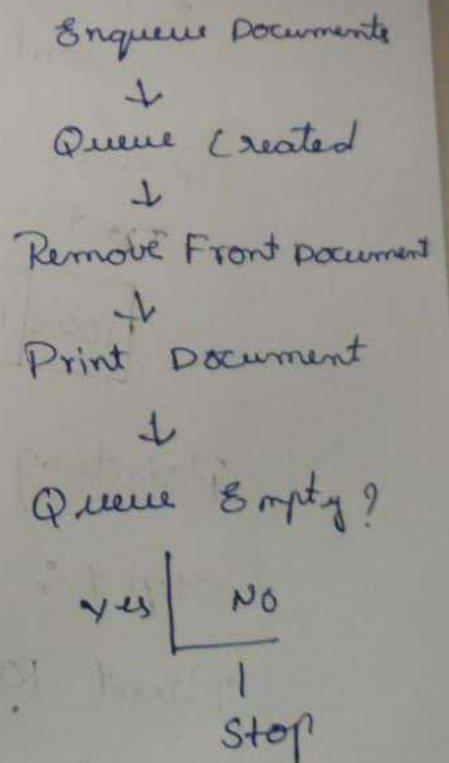
Step 6:

Print 1004

Queue becomes empty

Front = Rear = Null

Flow of printing



Algorithm :-

START

create an empty queue

Insert all documents into the queue

WHILE queue is not empty

Remove document from front

Print the document

END while

STOP

Time complexity

- Insert (Enqueue): $O(1)$
- Delete (Dequeue): $O(1)$
- Printing all documents: $O(n)$

2) Placement Training Register Number Search Problem

Find Register Number:

2014 2010

List (Unsorted):

2014 2015

2014 2033

2014 2011

2014 2017

2014 2016

20142056
20142003

Suitable Searching Technique

Linear Search

Reason

- The list is not sorted
- Linear search checks each element one by one

Searching Steps

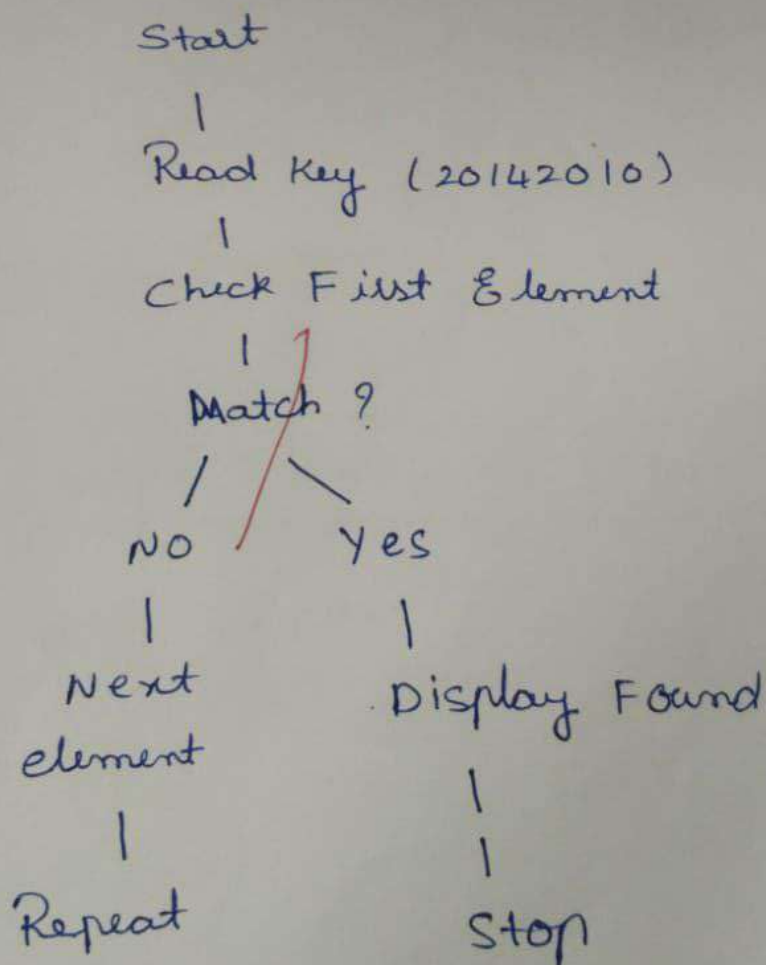
Step	Register Number	Match?
1	20142015	NO
	20142033	NO
2	20142011	NO
3	20142017	NO
4	20142016	Yes ✓
5		

The register number is found at position 5.

Procedure (Algorithm)

```
START
Read the register number to search
FOR each element in the list
    IF element == register number
        Display "Found"
        Stop
    END FOR
Display "NOT Found"
STOP
```

Flow of Linear Search



Time complexity

- Best case : $O(1)$ (Found at first position)
- Worst case : $O(n)$ (Found at last position or not found)

~~Worst~~